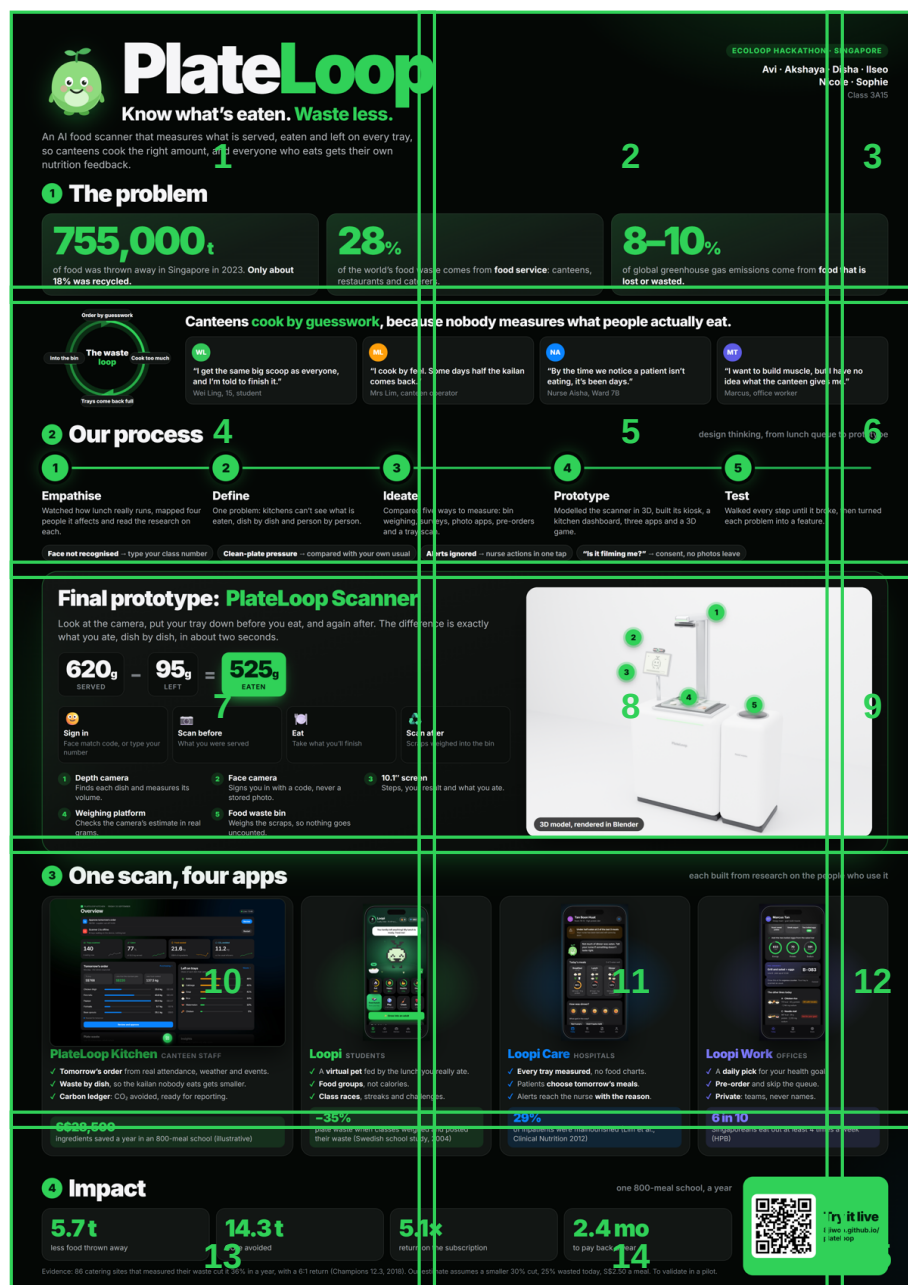


PlateLoop poster: A1 from 15 A4 sheets (home printer)

Print every sheet on A4 at 100% (Actual size, not "Fit"), landscape. Any printer works.

1. Lay the sheets out as numbered below: 3 across, 5 down.
2. On every sheet except the left column, cut along the LEFT crop marks.
3. On every sheet except the top row, cut along the TOP crop marks.
4. Slide each cut edge over its neighbour's 10 mm overlap until the picture lines up, then tape it from the back.
5. Trim the white margins around the outside last. The finished poster is exactly 594 x 841 mm.





PlateLoop

Know what's eaten. Waste less.

An AI food scanner that measures what is served, eaten and left on every tray, so canteens cook the right amount, and everyone who eats gets their own nutrition feedback.

1 The problem

755,000_t

of food was thrown away in Singapore in 2023. **Only about 18% was recycled.**

28%

of the world's food waste restaurants and caterers

ECOLOOP HACKATHON · SI

Avi · Akshaya · Di
Nicol

waste comes from **food service**: canteens,
erers.

8–10%

of global greenhouse gas emissions come from **food t**
lost or wasted.

• **SINGAPORE**

• **Disha** • **Ilseo**
nicole • **Sophie**

Class 3A15

od that is



Canteens **cook by guesswork**, because

WL

"I get the same big scoop as everyone, and I'm told to finish it."

Wei Ling, 15, student

ML

"I cook by feel. So much food comes back."

Mrs Lim, canteen owner

2 Our process

1

Empathise

Watched how lunch really runs, mapped four people it affects and read the research on each.

2

Define

One problem: kitchens can't see what is eaten, dish by dish and person by person.

3

Ideate

Compared five different weighing, surveying and a tray scanner.

Face not recognised → type your class number

Clean-plate pressure → compared with your own usual

All

use nobody measures what people actually eat.

el. Some days half the kailan
"
en operator

NA

"By the time we notice a patient isn't eating, it's been days."
Nurse Aisha, Ward 7B

MT

"I want to build muscle, but I ha
idea what the canteen gives me
Marcus, office worker

design thinking, from lunch queue to



Prototype

Modelled the scanner in 3D, built its kiosk, a kitchen dashboard, three apps and a 3D game.

d five ways to measure: bin
surveys, photo apps, pre-orders
scan.

Alerts ignored → nurse actions in one tap



Test

Walked every step until it broke, t
each problem into a feature.

"Is it filming me?" → consent, no photos leave

I have no
s me."

ue to prototype

ke, then turned

Final prototype: **PlateLoop Scanner**

Look at the camera, put your tray down before you eat, and again after. The difference is what you ate, dish by dish, in about two seconds.

620_g
SERVED

—

95_g
LEFT

=

525_g
EATEN



Sign in

Face match code, or type your number



Scan before

What you were served



Eat

Take what you'll finish



Scan after

Scraps w

1 Depth camera

Finds each dish and measures its volume.

2 Face camera

Signs you in with a code, never a stored photo.

3 10.1" screen

Steps, your rest

4 Weighing platform

Checks the camera's estimate in real grams.

5 Food waste bin

Weighs the scraps, so nothing goes uncoun

reference is exactly

on after

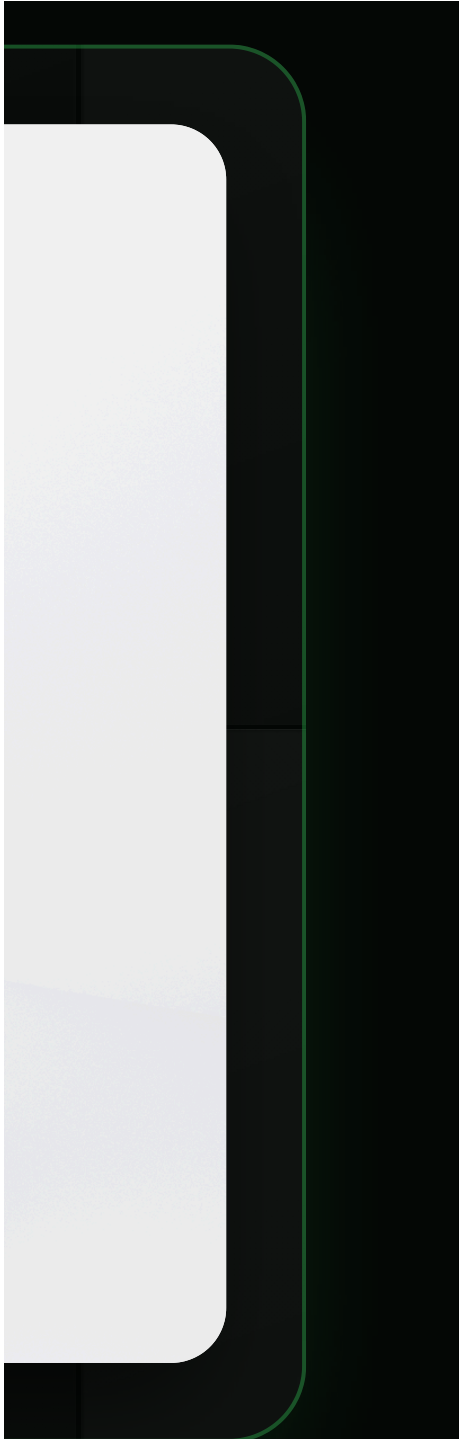
aps weighed into the bin

en

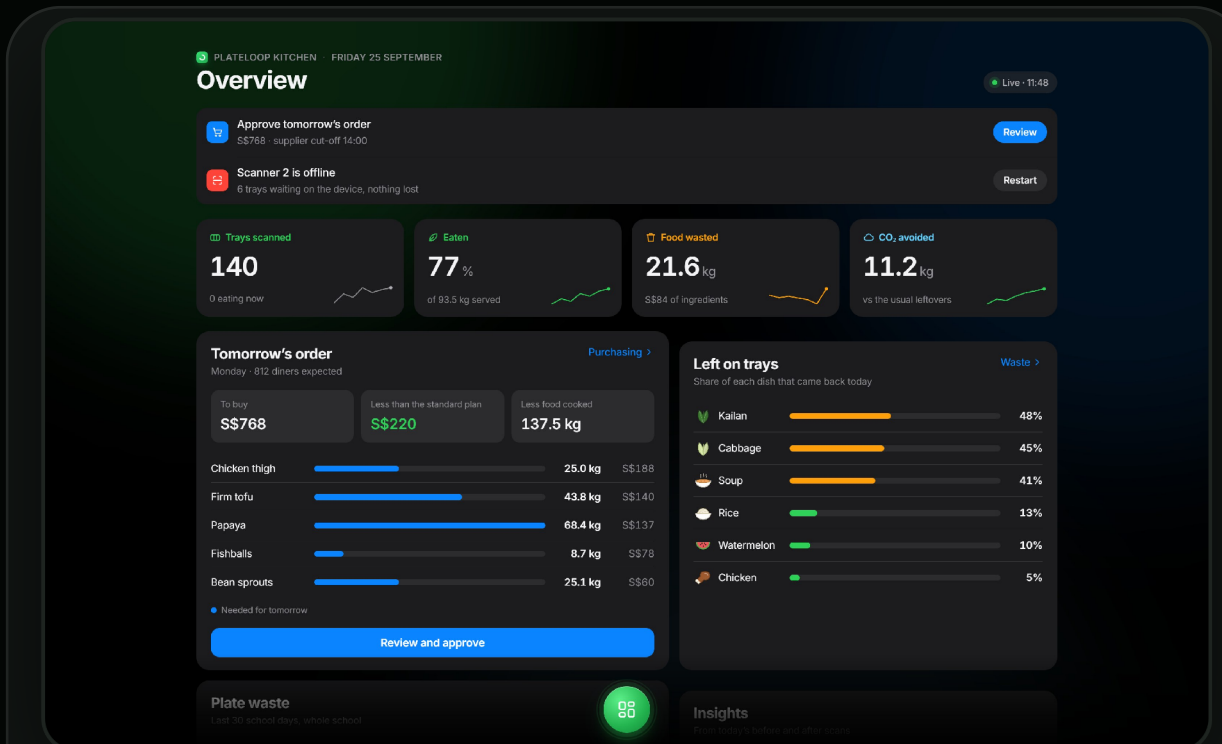
r result and what you ate.



3D model, rendered in Blender



3 One scan, four apps



PlateLoop Kitchen CANTEEN STAFF

- ✓ **Tomorrow's order** from real attendance, weather and events.
- ✓ **Waste by dish**, so the kailan nobody eats gets smaller.
- ✓ **Carbon ledger**: CO₂ avoided, ready for reporting.

S\$28 500

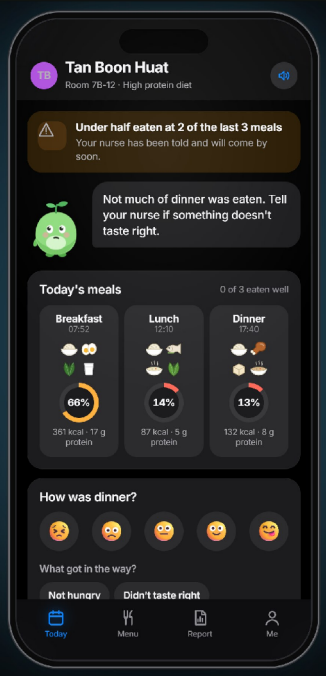


Loopi STUDENTS

- ✓ A **virtual pet** fed by the lunch you
- ✓ **Food groups**, not calories.
- ✓ **Class races**, streaks and challenges

-35%

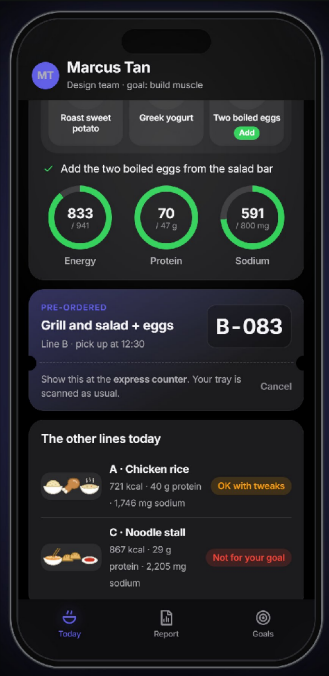
each built from research on the people



Loopi Care HOSPITALS

- ✓ Every tray measured, no food charts.
- ✓ Patients choose tomorrow's meals.
- ✓ Alerts reach the nurse with the reason.

29%



Loopi Work OFFICES

- ✓ A daily pick for your health goal.
- ✓ Pre-order and skip the queue.
- ✓ Private: teams, never names.

6 in 10



\$28,500

ingredients saved a year in an 800-meal school (illustrative)

-35%

plate waste when classes weighed and
their waste (Swedish school study, 2018)

4 Impact

5.7t

less food thrown away

14.3t

CO₂e avoided

5.1x

return on investment

Evidence: 86 catering sites that measured their waste cut it 36% in a year, with a 6:1 return (Champions 12.3, 2018). Our estimate is based on 100% waste reduction.

29%

of inpatients were malnourished (Lim et al.,
Clinical Nutrition 2012)

6 in 10

Singaporeans eat out at least 4 times a
(HPB)

hed and posted
dy, 2004)

one 800-meal school, a year

1x

on the subscription

2.4 mo

to pay back a year



Try
8jiwo
plate

estimate assumes a smaller 30% cut, 25% wasted today, S\$2.50 a meal. To validate in a pilot.

es a week

Try it live

[gijwoo.github.io/
plateloop](https://gijwoo.github.io/plateloop)